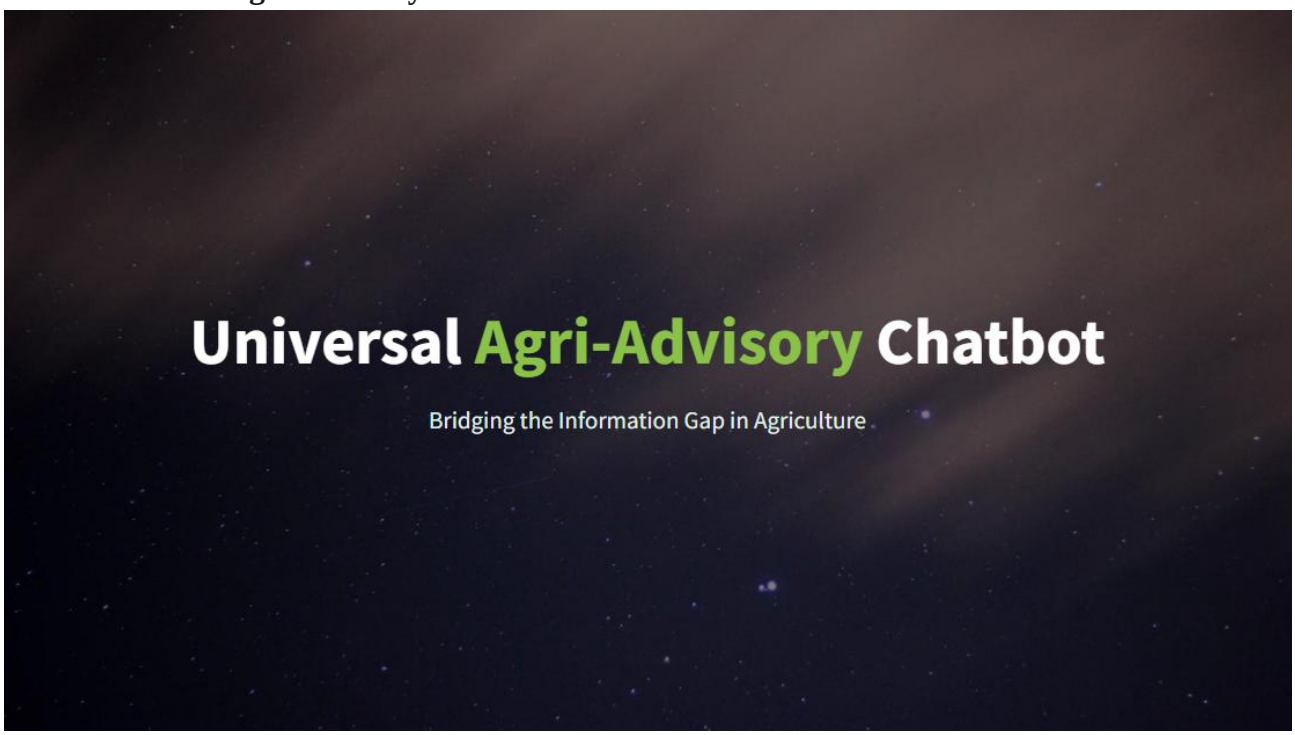
Universal Agri-Advisory Chatbot

1. Title: Universal Agri-Advisory Chatbot



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2. Project Statement:

Globally, farmers face challenges in accessing timely and accurate information regarding crop management, pest control, and optimal agricultural practices. This information gap often leads to suboptimal yields, increased costs, and environmental concerns due to inappropriate use of resources. Many existing solutions are language-specific or require advanced digital literacy. This project aims to bridge this gap by providing an accessible, **multi-language AI chatbot** that offers instant, reliable agricultural advice. The application will empower farmers with knowledge, helping them make informed decisions to improve productivity and sustainability.

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3. Outcomes:

- **Accessible agricultural advice** available in multiple languages (starting with English and Hindi).
- **Accurate and context-aware responses** to farmer queries on topics like crop diseases, pest identification, fertilizer use, and general farming practices.
- **Reduced information asymmetry** for farmers, leading to better decision-making.
- **Scalable knowledge base** architecture for easy expansion to more crops, regions, and languages.
- **Secure user management** for personalized interaction and feedback.

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4. Modules to be implemented:

- **User Authentication and Profile Management:** [1]
    1. User registration, login (standard email/password, with potential for future social logins). [2]
    2. Role-based authentication (farmer/admin). [3]
    3. User profile for basic agricultural context (e.g., primary crops grown, region, preferred language). [4]
- **Conversational AI Core (Chatbot Engine):**
    1. Natural Language Understanding (NLU) to interpret farmer queries.
    2. Dialogue Management to maintain conversation flow and context.
    3. Natural Language Generation (NLG) for providing accurate and helpful responses.
    4. Multi-language support for input and output.
- **Agricultural Knowledge Base Module:**
    1. Systematic storage and retrieval of agricultural data (crop information, pest details, disease symptoms, treatment options, fertilizer guidelines).
    2. Integration with open-source data sources or simulated data for weather/market prices.
- **Admin Dashboard & Content Management:**
    1. Admin interface for managing the agricultural knowledge base (adding/editing facts).
    2. Reviewing user interactions and identifying areas for chatbot improvement.
    3. User management and monitoring.

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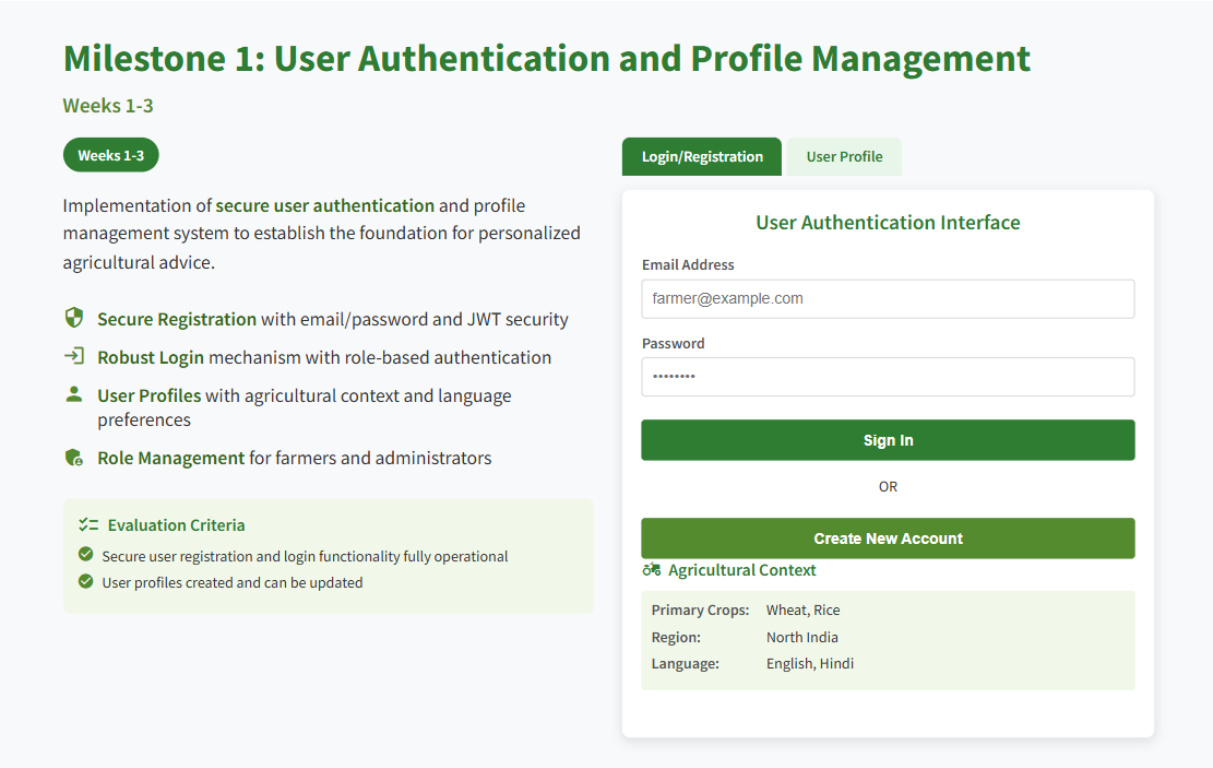
5. Week-wise module implementation and high-level requirements with output screenshots:

Milestone 1: Weeks 1-3

Module 1: User Authentication and Profile Management [5]

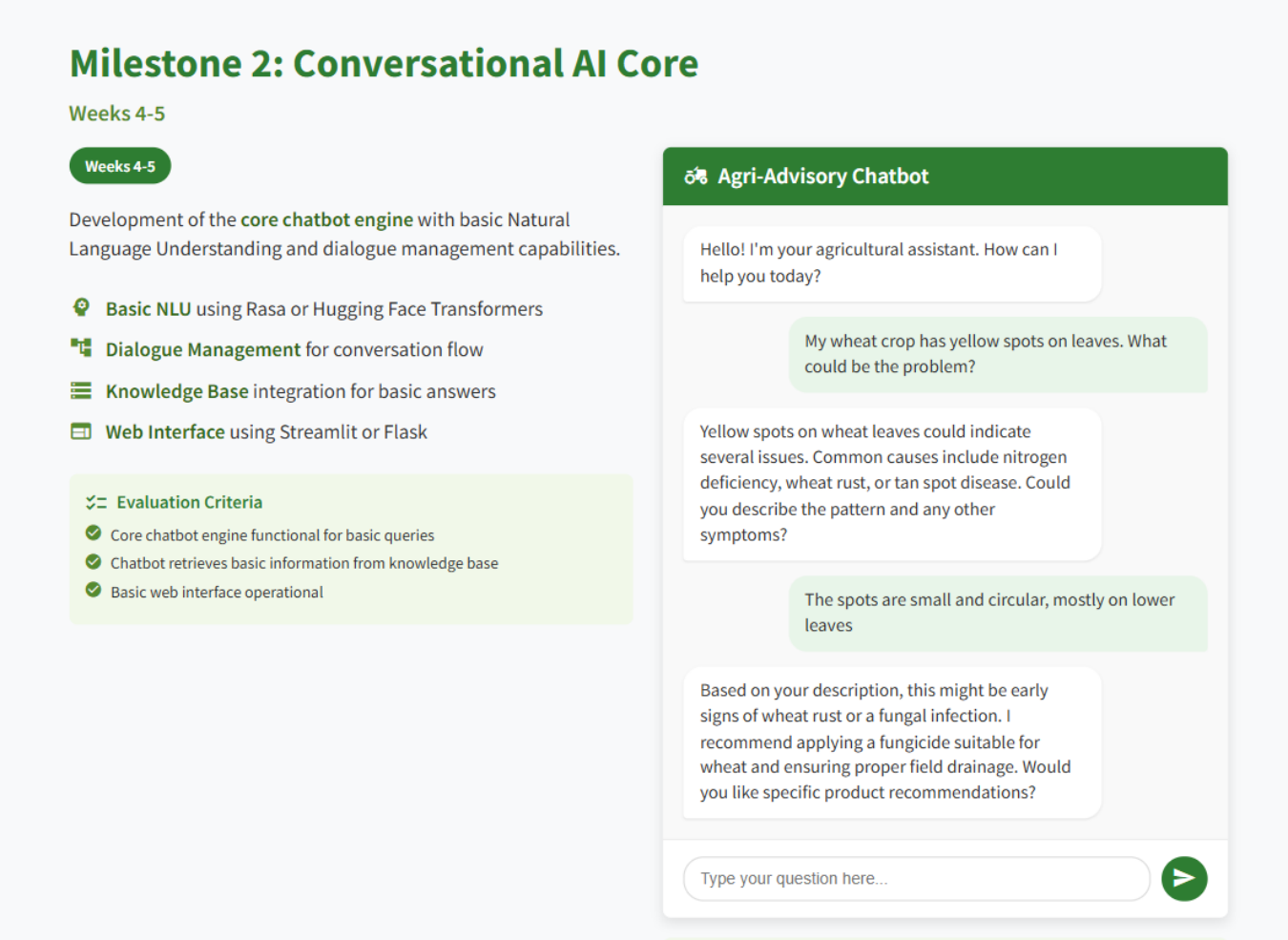
1. **User Registration:** Implement a secure user registration system with standard email/password and **JWT (JSON Web Token)** security for authentication.
2. **Login System:** Develop a robust login mechanism to authenticate users.
3. **Profile Management:** Create user profiles allowing farmers to update personal information and specify their primary crop interests and preferred language. This data will be used to personalize future interactions.



Milestone 2: Weeks 4-5

Module 2: Conversational AI Core (Chatbot Engine - Basic NLU & Dialogue Flow)

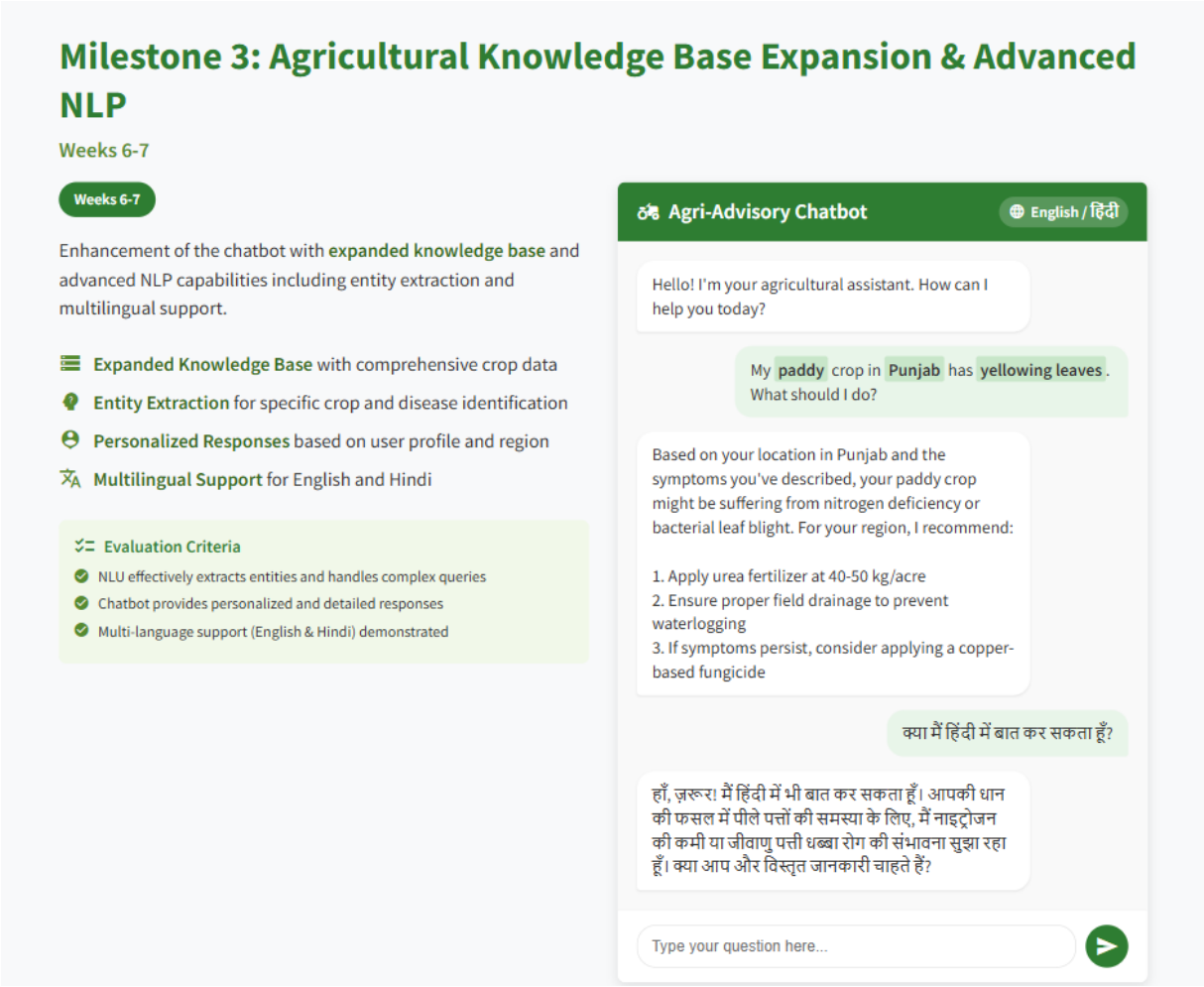
1. **Basic NLU & Intent Recognition:** Develop the core Natural Language Understanding component using **Rasa NLU** or fine-tuned **Hugging Face Transformers** to identify user intents.
2. **Simple Dialogue Management:** Implement basic dialogue flows using **Rasa Core** or a custom state machine to handle common farmer queries (e.g., asking for symptoms, requesting solutions).
3. **Initial Knowledge Base Integration:** Connect the chatbot to a small, pre-populated agricultural knowledge base (e.g., a JSON file or SQLite database) to retrieve basic answers.
4. **Web Interface (Initial Chat UI):** Design a simple web-based chat interface using **Streamlit** or a basic **Flask** app for user interaction.



Milestone 3: Weeks 6-7

Module 3: Agricultural Knowledge Base Expansion & Advanced NLP

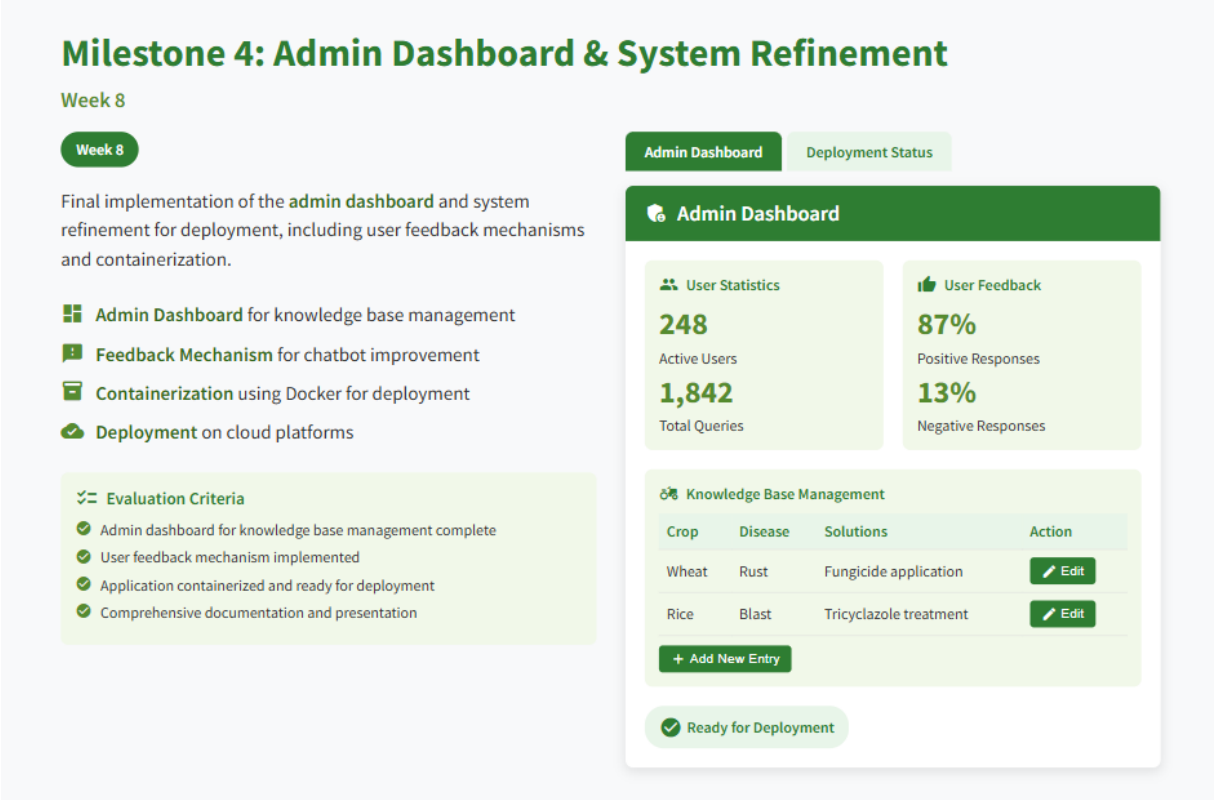
1. **Expanded Knowledge Base:** Populate the agricultural knowledge base with more comprehensive data on specific crops, common pests, and diseases (simulated or publicly available data from agricultural research institutes).
2. **Entity Extraction & Slot Filling:** Enhance NLU to extract specific entities from user queries (e.g., crop name "paddy", disease "blast").
3. **Response Generation & Personalization:** Improve Natural Language Generation to provide more detailed and personalized responses based on extracted entities and user profile information (e.g., "For *paddy blast* in your region...").
4. **Multilingual Expansion (Hindi Integration):** Implement initial support for Hindi language input and output using suitable multilingual models from **Hugging Face Transformers**.



Milestone 4: Weeks 8

Module 4: Admin Dashboard & System Refinement

1. **Admin Dashboard:** Develop a basic admin interface (using **Flask/Dash**) for managing the agricultural knowledge base (add/edit/delete crop data, disease info) and monitoring chatbot usage statistics.
2. **Feedback Mechanism:** Implement a simple user feedback system (e.g., thumbs up/down, brief comment) for chatbot responses.
3. **Deployment Preparation:** Containerize the application using **Docker**. Prepare for deployment on free cloud platforms (e.g., **Hugging Face Spaces, Streamlit Community Cloud**).
4. **Documentation & Presentation:** Finalize project documentation, code comments, and prepare for the final presentation.



6. Evaluation Criteria:

- **Milestone 1 Evaluation (End of Week 3):** [9]
  1. Secure user registration and login functionality is fully operational. [10]
  2. User profiles are created and can be updated.
- **Milestone 2 Evaluation (End of Week 5):** [11]
  1. Core chatbot engine (NLU, dialogue management) is functional for basic queries.
  2. Chatbot can retrieve and display basic information from the initial knowledge base.
  3. Basic web interface is operational.
- **Milestone 3 Evaluation (End of Week 7):** [12]
  1. NLU effectively extracts entities and handles complex queries.
  2. Chatbot provides personalized and detailed responses from an expanded knowledge base.
  3. Multi-language support (English & Hindi) for text input/output is demonstrated.
- **Milestone 4 Evaluation (End of Week 8):** [13]
  1. Admin dashboard for knowledge base management and user monitoring is complete. [14]
  2. User feedback mechanism is implemented.
  3. Application is containerized and ready for public deployment.
  4. Comprehensive documentation and successful final presentation.

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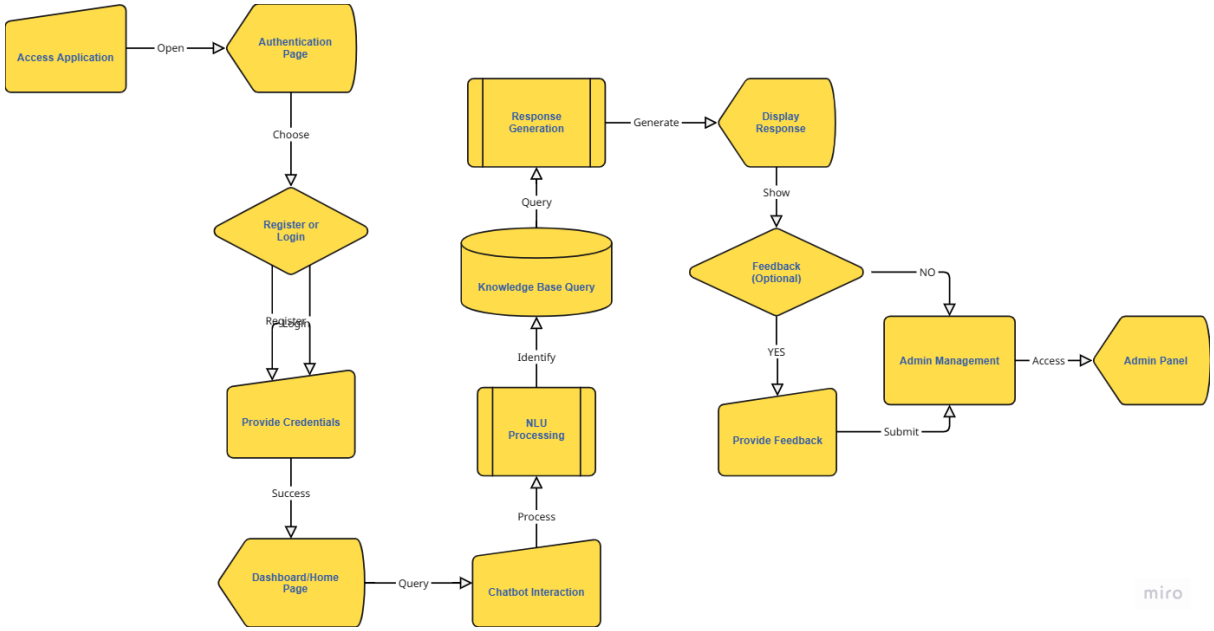
7. Workflow Diagram



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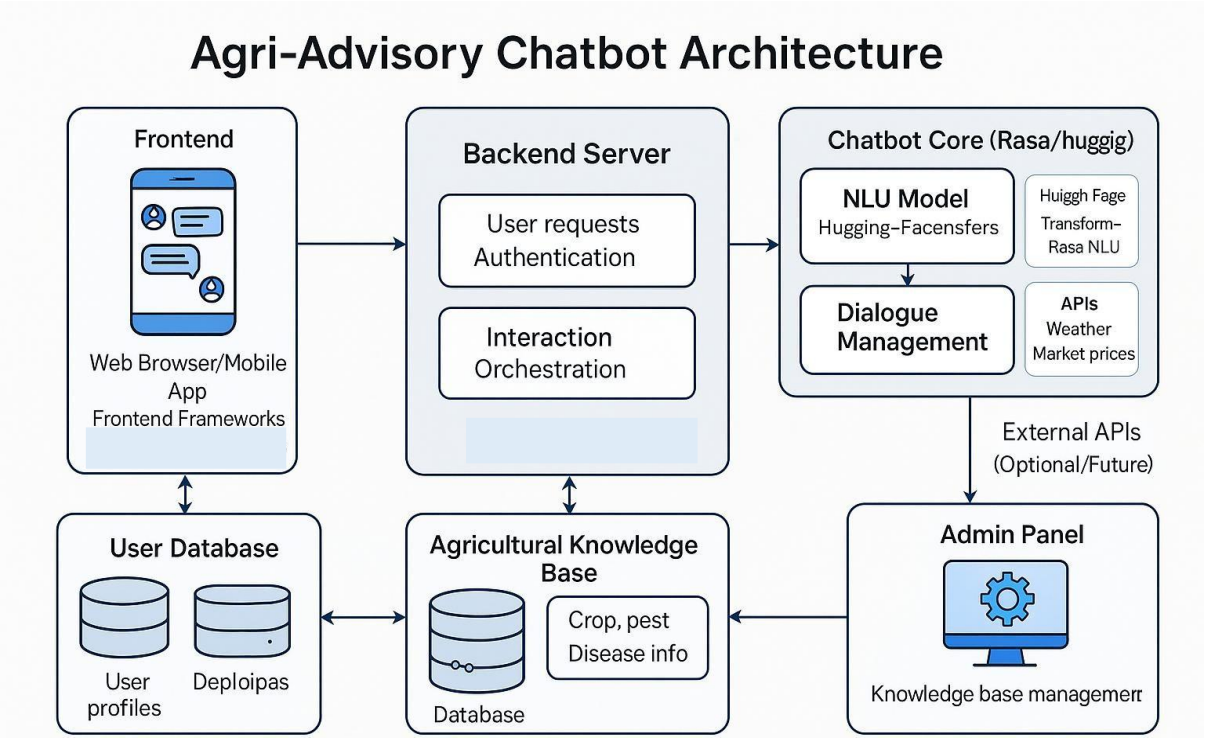
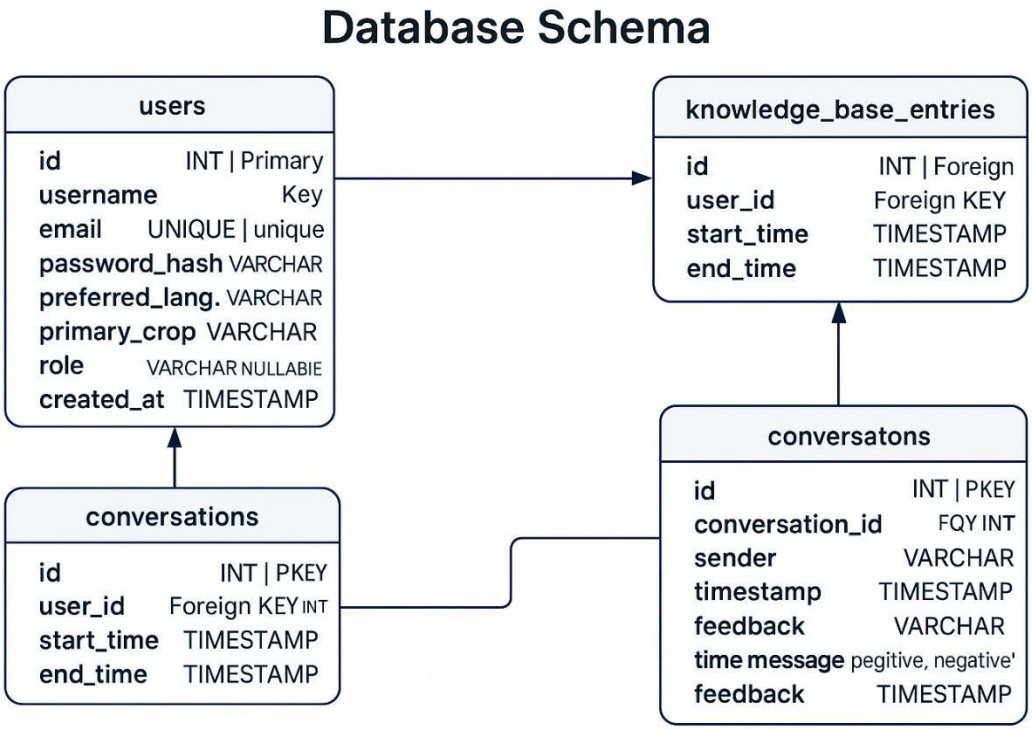
8. Architecture Diagram



9. Database Schema



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